

Codetantra Practical 4

Practice Lab Assignment

The screenshot displays the Codetantra web application interface. The top navigation bar includes the Codetantra logo, a search bar, and user account information. The main content area is titled "4.1.1. Pandas - series creation and manipulation". It contains a description of the task: "Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations." Below this, there are sections for "Input Format" and "Output Format". The "Input Format" section states: "The user should enter a list of numbers separated by space when prompted." The "Output Format" section states: "The program should display the mean of even and odd numbers separately. Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers." A "Sample Test Cases" section follows, showing two test cases. Test case 1 has input "1 2 3 4 5 6 7 8 9 10" and output: "Mean of even and odd numbers: 5.0", "Odd: 5.0", "Even: 6.0", "dtype: float64". Test case 2 has input "4 4 4 6 2 2 2 10 12 14 16" and output: "Mean of even and odd numbers: 6.333333", "Even: 6.333333", "dtype: float64". On the right side, there is a code editor with a Python script that implements the task. The script takes user input, creates a Pandas series, groups the data by even and odd numbers, and calculates the mean for each group. Below the code editor, there is a "Test cases" section showing the execution results. It indicates that 3 out of 3 shown test case(s) passed and 3 out of 3 hidden test case(s) passed. The "Test case 1" section shows the expected output and the actual output, which match.

Course

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4.1.2. Dictionary to dataframe

4.1.3. Student Information

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4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

Test case 1

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers: 5.0

Odd: 5.0

Even: 6.0

dtype: float64

Test case 2

4 4 4 6 2 2 2 10 12 14 16

Mean of even and odd numbers: 6.333333

Even: 6.333333

dtype: float64

seriesMa...

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5 # Create a Pandas series from the list of numbers
6 Series=pd.Series(numbers)
7
8 # Grouping by even and odd numbers and calculating the mean
9 grouped =Series.groupby(Series % 2 ==0).mean()
10
11 # Display the mean of even and odd numbers with labels
12 grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
13 print("Mean of even and odd numbers:")
14 print(grouped)
15
```

Average time 0.015 s Maximum time 0.043 s

14.83 ms 43.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 43 ms Debug

Expected output 1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers: 5.0

Odd: 5.0

Even: 6.0

dtype: float64

Actual output 1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers: 5.0

Odd: 5.0

Even: 6.0

dtype: float64

Terminal Test cases

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4.1.2 Dictionary to dataframe

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A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user.
- Display the DataFrame after adding the new column.

Delete a column:

- Remove the Age column.
- Display the DataFrame after deleting the column.

Sample Test Cases

Test case 1

original dataframe:

```
-----
name  age
0  Alice  25
1  Bob    30
2  Charlie 35
```

New name:

New age:

After adding a new row:

```
-----
name  age
0  Alice  25
1  Bob    30
2  Charlie 35
3  Alice  28
```

dataframe...

```
1 import pandas as pd
2
3 # Provided dictionary of lists
4 data = {
5     'name': ['Alice', 'Bob', 'Charlie'],
6     'age': [25, 30, 35]
7 }
8
9 # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11
12 # Display the original DataFrame
13 print("Original DataFrame:")
14 print(df)
15
16 # Adding a new row
17 new_name = input("New name: ")
18 new_age = int(input("New age: "))
19 df.loc[len(df)] = [new_name, new_age]
20
21 # Display the DataFrame after adding a new row
22 print("After adding a new row:")
23 print(df)
24
25 # Modifying a row
26 row_to_modify = int(input("Index of row to modify: "))
27 new_age_value = int(input("New age: "))
28 df.at[row_to_modify, 'age'] = new_age_value
29
30 # Display the DataFrame after modifying a row
31 print("After modifying a row:")
32 print(df)
33
34 # Deleting a row
35 row_to_delete = int(input("Index of row to delete: "))
36 df.drop(index=row_to_delete, reset_index(drop=True))
37 # Display the DataFrame after deleting a row
38 print("After deleting a row:")
39 print(df)
40
41 # Adding a new column
42 genders = input("Enter genders separated by space: ").split()
43 df['gender'] = genders
44
45 # Display the DataFrame after adding a new column
46 print("After adding a new column:")
47 print(df)
```

Execution time: 6.126 s
Execution time: 0.166 s

1 out of 1 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1 ☒ ☒ ☒

Expected output

Original dataframe:

Actual output:

name age

0 Alice 25

1 Bob 30

2 Charlie 35

3 Alice 28

4 Test Case

Course

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4.1.3. Student Information

2024

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

• Display the first five rows of the data frame.

• Calculate the average age of the students (limit the average age up to 2 decimal places).

• Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

Sample Test Cases

Test case 1

studentdata.txt

First five rows:

... Name Age Grade

0 John 25 A

1 Ailin 22 B

2 Emma 24 A

3 Mike 23 C

4 Sony 21 B

Average age: 23.29

Students with a grade up to B

... Name Age Grade

0 John 25 A

studentin...

Submit

1 import pandas as pd

2

3 # Read the text file into a DataFrame

4 file = input()

5 data = pd.read_csv(file, sep="\\s+", header=None, names=["Name", "Age", "Grade"])

6 print("First five rows:")

7 print(data.head())

8 aver=round(data["Age"].mean(),2)

9 print("Average age:",ave)

10 filtered_data=data[data["Grade"].isin(['A','B'])]

11 print("Students with a grade up to B")

12 print(filtered_data)

13

14

Average time 0.069 s 68.50 ms Maximum time 0.100 s 100.00 ms 1 out of 1 shown test case(s) passed 1 out of 1 hidden test case(s) passed

Test case 1 4000 ms Debug

Expected output Actual output

studentdata.txt studentdata.txt

First five rows:

... Name Age Grade

0 John 25 A

1 Ailin 22 B

2 Emma 24 A

Terminal

Test cases

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Lab Assignment

The screenshot displays a web-based coding environment with a sidebar on the left containing a course menu. The main area is divided into three sections: a problem description, sample data, and a code editor with a terminal and test cases.

Course Menu (Left Sidebar):

- Essentials of Data Science Laboratory - 2364152L - 2026
 - 3.2 Lab-Assignment
 - 3.2.1. Numpy: Matrix Operations
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 - 5. Practical 5

Problem Description (4.2.1. Month with the Highest Total Sales):

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
date,product,quantity,price,city
2023-01-01,Product A,1,20,New York
2023-01-05,Product B,2,15,Los Angeles
2023-01-02,Product A,7,30,New York
2023-01-03,Product C,4,30,Chicago
2023-01-05,Product B,2,15,Chicago
2023-01-03,Product A,8,30,Los Angeles
2023-01-04,Product C,5,30,New York
2023-01-06,Product B,5,15,Los Angeles
2023-01-01,Product A,3,30,Chicago
2023-01-05,Product C,10,30,Los Angeles
```

Note: The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases:

Test case 1

```
order_data.csv
Best month: 2023-01
Total sales: $1230.00
```

Code Editor:

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 # Convert Date column to datetime
10 df['Date'] = pd.to_datetime(df['Date'])
11
12 # Create Month column (format: YYYY-MM)
13 df['Month'] = df['Date'].dt.to_period('M').astype(str)
14
15 # Calculate Sales
16 df['Sales'] = df['Quantity'] * df['Price']
17
18 # Group by Month and sum sales
19 monthly_sales = df.groupby('Month')['Sales'].sum()
20
21 # Find best month
22 best_month = monthly_sales.idxmax()
23 highest_sales = monthly_sales.max()
24
25 print(f"Best month: {best_month}")
26 print(f"Total sales: ${highest_sales:.2f}")
27
```

Terminal:

```
44.67 ms 6.102 s 102.00 ms
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
```

Test Case 1:

Expected output	Actual output
order_data.csv	order_data.csv
Best month: 2023-01	Best month: 2023-01
Total sales: \$1230.00	Total sales: \$1230.00

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3.2 Lab Assignment

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4.2.2 Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2023-01-01	Product A	5	20	New York
2023-01-01	Product B	3	15	Los Angeles
2023-01-01	Product A	7	20	New York
2023-01-02	Product C	6	30	Chicago
2023-01-03	Product B	2	15	Chicago
2023-01-03	Product A	8	20	Los Angeles
2023-01-04	Product C	6	30	New York
2023-01-04	Product B	5	15	Los Angeles
2023-01-05	Product A	3	20	Chicago
2023-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

Test case 1

Expected output	Actual output
Best-selling product: Product A	Best-selling product: Product A
Total quantity sold: 23	Total quantity sold: 23

monthFor...

1import pandas as pd

2

3# Prompt the user for the file name

4file_name = input()

5

6# Load the data

7df = pd.read_csv(file_name)

8

9product_sales = df.groupby('Product')['Quantity'].sum()

10# Find the product with the highest total quantity sold

11best_product = product_sales.idxmax()

12highest_quantity = product_sales.max()

13

14# Display the result

15print(f"Best-selling product: {best_product}")

16print(f"Total quantity sold: {highest_quantity}")

17

Average time0.034 s34.00 msMaximum time0.078 s78.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test Case 16/8

Expected output

Actual output

Best-selling product: Product A

Best-selling product: Product A

Total quantity sold: 23

Total quantity sold: 23

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4.2.3. City that Sold the Most Products

05:51

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by city and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

Test case 1
sales_data.csv
City sold the most products: Los Angeles

monthFor...

Submit

1import pandas as pd
2
3# Prompt the user for the file name
4file_name = input()
5
6# Load the data
7df = pd.read_csv(file_name)
8
9# write the code..
10product_name=df.groupby('City')['Quantity'].sum()
11best_city=product_name.idxmax()
12# Display the result
13print(f"City sold the most products: {best_city}")
14

Average time0.035 s35.33 msMaximum time0.079 s79.00 ms1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 169 msDebugTest cases
Expected outputActual output
sales_data.csvsales_data.csv
City sold the most products: Los AngelesCity sold the most products: Los Angeles

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- 5 Practical 5

4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pairs that was sold most frequently.

Sample Data:

```
date,product,quantity,price,city
2025-01-01,Product A,5,30,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,30,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,4,30,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product A,5,15,Los Angeles
2025-01-05,Product A,3,30,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Explanation:
Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product B, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product A
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together.

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

Test case 1

Explorer

1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10 # Group products by Date (transactions)
11 grouped = df.groupby('Date')['Product'].apply(list)
12 # Generate pairs
13 pair_counts = Counter()
14 for products in grouped:
15 pairs = combinations(sorted(products), 2)
16 pair_counts.update(pairs)
17
18 # Find maximum frequency
19 max_count = max(pair_counts.values())
20 # Get most frequent pairs
21 most_common_pairs = [pair for pair, count in pair_counts.items() if count == max_count]
22
23 # Sort pairs alphabetically
24 most_common_pairs.sort()
25 # Output
26 for pair in most_common_pairs:
27 print(f"({pair[0]} and {pair[1]}): {max_count} times")

Average time: 0.021 s
Maximum time: 0.035 s
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1
Expected output: sales_data.csv
Actual output: sales_data.csv
Product A and Product B: 2 times
Product A and Product C: 2 times

Terminal Test cases

Course

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Search course

3.1 Practice Lab Assignment

3.1.1 List Operations

3.1.2 Dictionary Operations

3.2 Lab Assignment

3.2.1 Linear Search Technique

3.2.2 Captain of the Team

3.3 Practical 3

3.3.1 Practice Lab Assignment

3.3.1.1 NumPy Array Operations

3.3.2 Lab Assignment

3.3.2.1 NumPy Matrix Operations

3.3.2.2 NumPy Horizontal and Vertical Slicing

3.3.2.3 NumPy Custom Sequence Generation

3.3.2.4 NumPy Arithmetic and Statistical Operations

3.3.2.5 NumPy Clipping and Masking Arrays

3.3.2.6 NumPy Reshaping, Sorting, Counting

3.3.2.7 Student Data Analysis and Operations

3.4 Practical 4

3.4.1 Practice Lab Assignment

3.4.1.1 Pandas - series creation and manipulations

3.4.1.2 Dictionary to DataFrame

3.4.1.3 Student Information

3.4.2 Lab Assignment

3.4.2.1 Start with the Highest Test Date

3.4.2.2 Map Setting Product

3.4.2.3 City that Sold the Most Products

3.4.2.4 Most Frequently Sold Product Pairs

3.4.3 Titanic Dataset Analysis and Data Cleaning

3.4.3.1 Titanic Dataset Analysis and Data Cleaning

3.4.3.2 Titanic Dataset Analysis and Data Cleaning

3.4.3.3 Titanic Dataset Analysis and Data Cleaning

3.4.3.4 Titanic Dataset Analysis and Data Cleaning

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3.4.3.98 Titanic Dataset Analysis and Data Cleaning

3.4.3.99 Titanic Dataset Analysis and Data Cleaning

3.4.3.100 Titanic Dataset Analysis and Data Cleaning

3.4.3.1 Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.

2. Display the last 5 rows of the dataset.

3. Get the shape of the dataset (number of rows and columns).

4. Get a summary of the dataset using .info().

5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().

6. Check for missing values and display the count of missing values for each column.

7. Fill missing values in the Age column with the median age.

8. Fill missing values in the Embarked column with the most frequent value (mode).

9. Drop the Cabin column due to many missing values.

10. Create a new column, FamilySize, by adding the SibSp and Parch columns.

The Titanic dataset contains columns as shown below.

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Braund, Mr. Henry James	Male	22	1	0	1015168	5.12	NaN	S
2	1	1	Mr. Allen, Mr. William Charles	Male	35	1	0	15170	5.21	NaN	C
3	0	3	Mr. Miley, Mr. David Bruce	Male	19	0	0	195900	3.36	NaN	S
4	1	3	Mr. Fyfe, Mr. James	Male	33	0	0	174630	5.69	NaN	S
5	0	3	Mr. Carter, Mr. Thomas	Male	25	0	0	14950	5.21	NaN	S
6	0	3	Mr. Williams, Mr. John	Male	26	0	0	15170	5.21	NaN	C
7	0	3	Mr. Anderson, Mr. John	Male	26	0	0	15170	5.21	NaN	C
8	0	3	Mr. Taylor, Mr. John	Male	26	0	0	15170	5.21	NaN	C
9	0	3	Mr. Brown, Mr. John	Male	26	0	0	15170	5.21	NaN	C
10	0	3	Mr. Green, Mr. John	Male	26	0	0	15170	5.21	NaN	C
11	0	3	Mr. White, Mr. John	Male	26	0	0	15170	5.21	NaN	C
12	0	3	Mr. Black, Mr. John	Male	26	0	0	15170	5.21	NaN	C
13	0	3	Mr. Gray, Mr. John	Male	26	0	0	15170	5.21	NaN	C
14	0	3	Mr. King, Mr. John	Male	26	0	0	15170	5.21	NaN	C
15	0	3	Mr. Hall, Mr. John	Male	26	0	0	15170	5.21	NaN	C
16	0	3	Mr. Young, Mr. John	Male	26	0	0	15170	5.21	NaN	C
17	0	3	Mr. Evans, Mr. John	Male	26	0	0	15170	5.21	NaN	C
18	0	3	Mr. Roberts, Mr. John	Male	26	0	0	15170	5.21	NaN	C
19	0	3	Mr. Turner, Mr. John	Male	26	0	0	15170	5.21	NaN	C
20	0	3	Mr. Phillips, Mr. John	Male	26	0	0	15170	5.21	NaN	C
21	0	3	Mr. Parker, Mr. John	Male	26	0	0	15170	5.21	NaN	C
22	0	3	Mr. Scott, Mr. John	Male	26	0	0	15170	5.21	NaN	C
23	0	3	Mr. Green, Mr. John	Male	26	0	0	15170	5.21	NaN	C
24	0	3	Mr. White, Mr. John	Male	26	0	0	15170	5.21	NaN	C
25	0	3	Mr. Black, Mr. John	Male	26	0	0	15170	5.21	NaN	C
26	0	3	Mr. Gray, Mr. John	Male	26	0	0	15170	5.21	NaN	C
27	0	3	Mr. King, Mr. John	Male	26	0	0	15170	5.21	NaN	C
28	0	3	Mr. Hall, Mr. John	Male	26	0	0	15170	5.21	NaN	C
29	0	3	Mr. Young, Mr. John	Male	26	0	0	15170	5.21	NaN	C
30	0	3	Mr. Evans, Mr. John	Male	26	0	0	15170	5.21	NaN	C
31	0	3	Mr. Roberts, Mr. John	Male	26	0	0	15170	5.21	NaN	C
32	0	3	Mr. Turner, Mr. John	Male	26	0	0	15170	5.21	NaN	C
33	0	3	Mr. Phillips, Mr. John	Male	26	0	0	15170	5.21	NaN	C
34	0	3	Mr. Parker, Mr. John	Male	26	0	0	15170	5.21	NaN	C
35	0	3	Mr. Scott, Mr. John	Male	26	0	0	15170	5.21	NaN	C
36	0	3	Mr. Green, Mr. John	Male	26	0	0	15170	5.21	NaN	C
37	0	3	Mr. White, Mr. John	Male	26	0	0	15170	5.21	NaN	C
38	0	3	Mr. Black, Mr. John	Male	26	0	0	15170	5.21	NaN	C
39	0	3	Mr. Gray, Mr. John	Male	26	0	0	15170	5.21	NaN	C
40	0	3	Mr. King, Mr. John	Male	26	0	0	15170	5.21	NaN	C
41	0	3	Mr. Hall, Mr. John	Male	26	0	0	15170	5.21	NaN	C
42	0	3	Mr. Young, Mr. John	Male	26	0	0	15170	5.21	NaN	C
43	0	3	Mr. Evans, Mr. John	Male	26	0	0	15170	5.21	NaN	C
44	0	3	Mr. Roberts, Mr. John	Male	26	0	0	15170	5.21	NaN	C
45	0	3	Mr. Turner, Mr. John	Male	26	0	0	15170	5.21	NaN	C
46	0	3	Mr. Phillips, Mr. John	Male	26	0	0	15170	5.21	NaN	C
47	0	3	Mr. Parker, Mr. John	Male	26	0	0	15170	5.21	NaN	C
48	0	3	Mr. Scott, Mr. John	Male	26	0	0	15170	5.21	NaN	C
49	0	3	Mr. Green, Mr. John	Male	26	0	0	15170	5.21	NaN	C
50	0	3	Mr. White, Mr. John	Male	26	0	0	15170	5.21	NaN	C
51	0	3	Mr. Black, Mr. John	Male	26	0	0	15170	5.21	NaN	C
52	0	3	Mr. Gray, Mr. John	Male	26	0	0	15170	5.21	NaN	C
53	0	3	Mr. King, Mr. John	Male	26	0	0	15170	5.21	NaN	C
54	0	3	Mr. Hall, Mr. John	Male	26	0	0	15170	5.21	NaN	C
55	0	3	Mr. Young, Mr. John	Male	26	0	0	15170	5.21	NaN	C
56	0	3	Mr. Evans, Mr. John	Male	26	0	0	15170	5.21	NaN	C
57	0	3	Mr. Roberts, Mr. John	Male	26	0	0	15170	5.21	NaN	C
58	0	3	Mr. Turner, Mr. John	Male	26	0	0	15170	5.21	NaN	C
59	0	3	Mr. Phillips, Mr. John	Male	26	0	0	15170	5.21	NaN	C
60	0	3	Mr. Parker, Mr. John	Male	26	0	0	15170	5.21	NaN	C
61	0	3	Mr. Scott, Mr. John	Male	26	0	0	15170	5.21	NaN	C
62	0	3	Mr. Green, Mr. John	Male	26	0	0	15170	5.21	NaN	C
63	0	3	Mr. White, Mr. John	Male	26	0	0	15170	5.21	NaN	C
64	0	3	Mr. Black, Mr. John	Male	26	0	0	15170	5.21	NaN	C
65	0	3	Mr. Gray, Mr. John	Male	26	0	0	15170	5.21	NaN	C
66	0	3	Mr. King, Mr. John	Male	26	0	0	15170	5.21	NaN	C
67	0	3	Mr. Hall, Mr. John	Male	26	0	0	15170	5.21	NaN	C
68	0	3	Mr. Young, Mr. John	Male	26	0	0	15170	5.21	NaN	C
69	0	3	Mr. Evans, Mr. John	Male	26	0	0	15170	5.21	NaN	C
70	0	3	Mr. Roberts, Mr. John	Male	26	0	0	15170	5.21	NaN	C
71	0	3	Mr. Turner, Mr. John	Male	26	0	0	15170	5.21	NaN	C
72	0	3	Mr. Phillips, Mr. John	Male	26	0	0	15170	5.21	NaN	C
73	0	3	Mr. Parker, Mr. John	Male	26	0	0	15170	5.21	NaN	C
74	0	3	Mr. Scott, Mr. John	Male	26	0	0	15170	5.21	NaN	C
75	0	3	Mr. Green, Mr. John	Male	26	0	0	15170	5.21	NaN	C
76	0	3	Mr. White, Mr. John	Male	26	0	0	15170	5.21	NaN	C
77	0	3	Mr. Black, Mr. John	Male	26	0	0	15170	5.21	NaN	C
78	0	3	Mr. Gray, Mr. John	Male	26	0	0	15170	5.21	NaN	C
79	0	3	Mr. King, Mr. John	Male	26	0	0	15170	5.21	NaN	C
80	0	3	Mr. Hall, Mr. John	Male	26	0	0	15170	5.21	NaN	C
81	0	3	Mr. Young, Mr. John	Male	26	0	0	15170	5.21	NaN	C
82	0	3	Mr. Evans, Mr. John	Male	26	0	0	15170	5.21	NaN	C
83	0	3	Mr. Roberts, Mr. John	Male	26	0	0	15170	5.21	NaN	C
84	0	3	Mr. Turner, Mr. John	Male	26	0	0	15170	5.21	NaN	C
85	0	3	Mr. Phillips, Mr. John	Male	26	0	0	15170	5.21	NaN	C
86	0	3	Mr. Parker, Mr. John	Male	26	0	0	15170	5.21	NaN	C
87	0	3	Mr. Scott, Mr. John	Male	26	0	0	15170	5.21	NaN	C
88	0	3	Mr. Green, Mr. John	Male	26	0	0	15170	5.21	NaN	C
89	0	3	Mr. White, Mr. John	Male	26	0	0	15170	5.21	NaN	C
90	0	3	Mr. Black, Mr. John	Male	26	0	0	15170	5.21	NaN	C
91	0	3	Mr. Gray, Mr. John	Male	26	0	0	15170	5.21	NaN	C
92	0	3	Mr. King, Mr. John	Male	26	0	0	15170	5.21	NaN	C
93	0	3	Mr. Hall, Mr. John	Male	26	0	0	15170	5.21	NaN	C
94	0	3	Mr. Young, Mr. John	Male	26	0	0	15170	5.21	NaN	C
95	0	3	Mr. Evans, Mr. John	Male	26	0	0	15170	5.21	NaN	C
96	0	3	Mr. Roberts, Mr. John	Male	26	0	0	15170	5.21	NaN	C
97	0	3	Mr. Turner, Mr. John	Male	26	0	0	15170	5.21	NaN	C
98	0	3	Mr. Phillips, Mr. John	Male	26	0	0	15170	5.21	NaN	C
99	0	3	Mr. Parker, Mr. John	Male	26	0	0	15170	5.21	NaN	C
100	0	3	Mr. Scott, Mr. John	Male	26	0	0	15170	5.21	NaN	C

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Mr. Braund, Mr. Henry James", Male, 22, 1, 0, 1015168, 5.12, NaN, S
2, 1, 1, "Mr. Allen, Mr. William Charles", Male, 35, 1, 0, 15170, 5.21, NaN, C
3, 0, 3, "Mr. Miley, Mr. David Bruce", Male, 19, 0, 0, 195900, 3.36, NaN, S
4, 1, 3, "Mr. Fyfe, Mr. James", Male, 33, 0, 0, 174630, 5.69, NaN, S
5, 0, 3, "Mr. Carter, Mr. Thomas", Male, 25, 0, 0, 14950, 5.21, NaN, S
6, 0, 3, "Mr. Williams, Mr. John", Male, 26, 0, 0, 15170, 5.21, NaN, C
7, 0, 3, "Mr. Anderson, Mr. John", Male, 26, 0, 0, 15170, 5.21, NaN, C
8, 0, 3, "Mr. Taylor, Mr. John", Male, 26, 0, 0, 15170, 5.21, NaN, C
9, 0, 3, "Mr. Brown, Mr. John", Male, 26, 0, 0, 15170, 5.21, NaN, C
10, 0, 3, "Mr. Green, Mr. John", Male, 26, 0, 0, 15170,
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Course

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4.2.8. Titanic Dataset Analysis and Data Cl...

5. Practical 5

4.2.7. Titanic Dataset Analysis and Data Cloning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.

2. Calculate the survival rate by embarkation location (Embarked_S).

3. Calculate the survival rate by family size (FamilySize).

4. Calculate the survival rate by being alone (alone).

5. Get the average fare by passenger class (Pclass).

6. Get the average age by passenger class (Pclass).

7. Get the average age by survival status (Survived).

8. Get the average fare by survival status (Survived).

9. Get the number of survivors by class (Pclass).

10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below.

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parth	Total	Fare	Cabin	Embarked
1	0	3	Mr. Braund, Mr. Owen	Male	22	1	0	21.0153	5.12	C101	S
2	1	1	Mr. Allen, Mr. William Henry	Male	29	0	0	51.0185	53.1	C103	S
3	0	3	Mr. McCarty, Mr. Timothy J	Male	34	1	1	51.0185	51.66	C101	S
4	1	1	Mr. Brown, Mr. John G	Male	54	1	1	51.0185	51.66	C101	S
5	0	3	Mr. Carter, Mr. James H	Male	35	1	1	51.0185	51.66	C101	S
6	1	1	Mr. Davis, Mr. John A	Male	50	1	1	51.0185	51.66	C101	S
7	0	3	Mr. Edwards, Mr. John D	Male	39	1	1	51.0185	51.66	C101	S
8	1	1	Mr. Fisher, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
9	0	3	Mr. Gibson, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
10	1	1	Mr. Harbeck, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
11	0	3	Mr. Hendon, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
12	1	1	Mr. Jones, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
13	0	3	Mr. Keith, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
14	1	1	Mr. Lester, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
15	0	3	Mr. Martin, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
16	1	1	Mr. Nash, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
17	0	3	Mr. O'Brien, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
18	1	1	Mr. Quinn, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
19	0	3	Mr. Ryan, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
20	1	1	Mr. Scott, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
21	0	3	Mr. Taylor, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
22	1	1	Mr. White, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
23	0	3	Mr. Black, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
24	1	1	Mr. Green, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
25	0	3	Mr. Hall, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
26	1	1	Mr. King, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
27	0	3	Mr. Long, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
28	1	1	Mr. Moore, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
29	0	3	Mr. Parker, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
30	1	1	Mr. Roberts, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
31	0	3	Mr. Turner, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
32	1	1	Mr. Walker, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
33	0	3	Mr. Young, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
34	1	1	Mr. Adams, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
35	0	3	Mr. Baker, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
36	1	1	Mr. Clark, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
37	0	3	Mr. Evans, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
38	1	1	Mr. Foster, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
39	0	3	Mr. Grant, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
40	1	1	Mr. Harris, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
41	0	3	Mr. Ives, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
42	1	1	Mr. Jordan, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
43	0	3	Mr. Kelly, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
44	1	1	Mr. Lester, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
45	0	3	Mr. Martin, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
46	1	1	Mr. Nash, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
47	0	3	Mr. O'Brien, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
48	1	1	Mr. Quinn, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
49	0	3	Mr. Ryan, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
50	1	1	Mr. Scott, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
51	0	3	Mr. Taylor, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
52	1	1	Mr. White, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
53	0	3	Mr. Black, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
54	1	1	Mr. Green, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
55	0	3	Mr. Hall, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
56	1	1	Mr. King, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
57	0	3	Mr. Long, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
58	1	1	Mr. Moore, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
59	0	3	Mr. Parker, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
60	1	1	Mr. Roberts, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
61	0	3	Mr. Turner, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
62	1	1	Mr. Walker, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
63	0	3	Mr. Young, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
64	1	1	Mr. Adams, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
65	0	3	Mr. Baker, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
66	1	1	Mr. Clark, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
67	0	3	Mr. Evans, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
68	1	1	Mr. Foster, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
69	0	3	Mr. Grant, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
70	1	1	Mr. Harris, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
71	0	3	Mr. Ives, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
72	1	1	Mr. Jordan, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
73	0	3	Mr. Kelly, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
74	1	1	Mr. Lester, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
75	0	3	Mr. Martin, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
76	1	1	Mr. Nash, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
77	0	3	Mr. O'Brien, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
78	1	1	Mr. Quinn, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
79	0	3	Mr. Ryan, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
80	1	1	Mr. Scott, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
81	0	3	Mr. Taylor, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
82	1	1	Mr. White, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
83	0	3	Mr. Black, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
84	1	1	Mr. Green, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
85	0	3	Mr. Hall, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
86	1	1	Mr. King, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
87	0	3	Mr. Long, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
88	1	1	Mr. Moore, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
89	0	3	Mr. Parker, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
90	1	1	Mr. Roberts, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
91	0	3	Mr. Turner, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
92	1	1	Mr. Walker, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
93	0	3	Mr. Young, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
94	1	1	Mr. Adams, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
95	0	3	Mr. Baker, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
96	1	1	Mr. Clark, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
97	0	3	Mr. Evans, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
98	1	1	Mr. Foster, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S
99	0	3	Mr. Grant, Mr. John A	Male	39	1	1	51.0185	51.66	C101	S
100	1	1	Mr. Harris, Mr. John A	Male	49	1	1	51.0185	51.66	C101	S

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parth, Total, Fare, Cabin, Embarked

1, 0, 3, Mr. Braund, Mr. Owen, 22, 1, 0, 21.0153, 5.12, C101, S

2, 1, 1, Mr. Allen, Mr. William Henry, 29, 0, 0, 51.0185, 53.1, C103, S

3, 0, 3, Mr. McCarty, Mr. Timothy J, 34, 1, 1, 51.0185, 51.66, C101, S

4, 1, 1, Mr. Brown, Mr. John G, 50, 1, 1, 51.0185, 51.66, C101, S

5, 0, 3, Mr. Carter, Mr. James H, 39, 1, 1, 51.0185, 51.66, C101, S

6, 1, 1, Mr. Davis, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

7, 0, 3, Mr. Edwards, Mr. John D, 39, 1, 1, 51.0185, 51.66, C101, S

8, 1, 1, Mr. Fisher, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

9, 0, 3, Mr. Gibson, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

10, 1, 1, Mr. Harbeck, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

11, 0, 3, Mr. Hendon, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

12, 1, 1, Mr. Jones, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

13, 0, 3, Mr. Keith, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

14, 1, 1, Mr. Lester, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

15, 0, 3, Mr. Martin, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

16, 1, 1, Mr. Nash, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

17, 0, 3, Mr. O'Brien, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

18, 1, 1, Mr. Quinn, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

19, 0, 3, Mr. Ryan, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

20, 1, 1, Mr. Scott, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

21, 0, 3, Mr. Taylor, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

22, 1, 1, Mr. White, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

23, 0, 3, Mr. Black, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

24, 1, 1, Mr. Green, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

25, 0, 3, Mr. Hall, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

26, 1, 1, Mr. King, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

27, 0, 3, Mr. Long, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

28, 1, 1, Mr. Moore, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

29, 0, 3, Mr. Parker, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

30, 1, 1, Mr. Roberts, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

31, 0, 3, Mr. Turner, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

32, 1, 1, Mr. Walker, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

33, 0, 3, Mr. Young, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

34, 1, 1, Mr. Adams, Mr. John A, 49, 1, 1, 51.0185, 51.66, C101, S

35, 0, 3, Mr. Baker, Mr. John A, 39, 1, 1, 51.0185, 51.66, C101, S

36, 1, 1, Mr. Clark, Mr. John A, 49,

